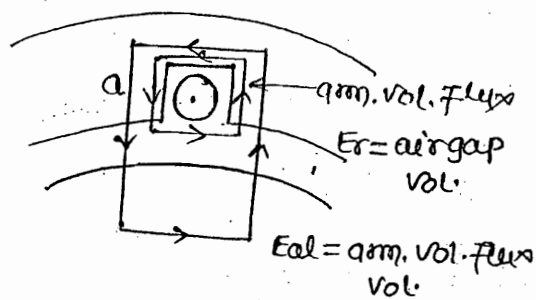
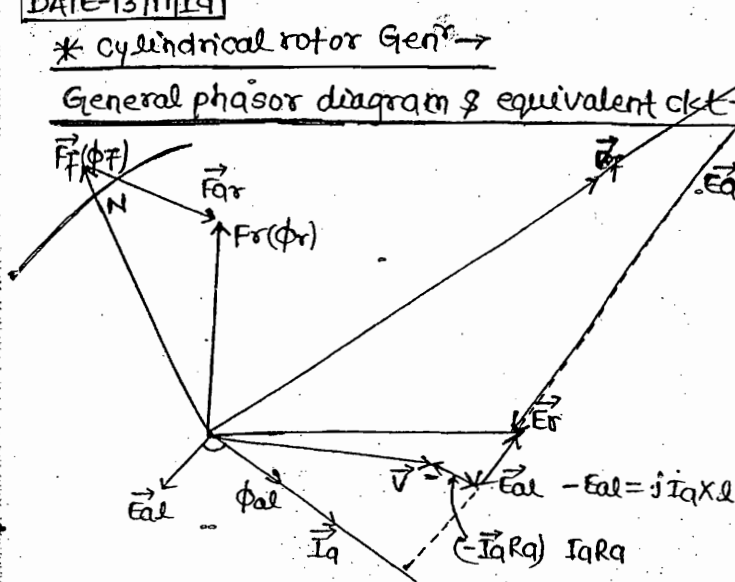


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* Cylindrical rotor Gen^r →

General phasor diagram & equivalent ckt →



E_f = excitation vol.
(OR)

Internal vol.
(OR)

OC voltage

E_{ar} = armature reaction vol.

X_l = Armature leakage reactance

X_{ar} = armature reaction reactance
(OR)

magnetising Reactance of
synchronous Re m/c

X_s = synchronous Reactance
($X_l + X_{ar}$)

$$\vec{V} = \vec{E}_r + \vec{E}_{al} - \vec{I}_a R_a$$

$$\vec{V} = (\vec{E}_f + \vec{E}_{ar}) + \vec{E}_{al} - \vec{I}_a R_a$$

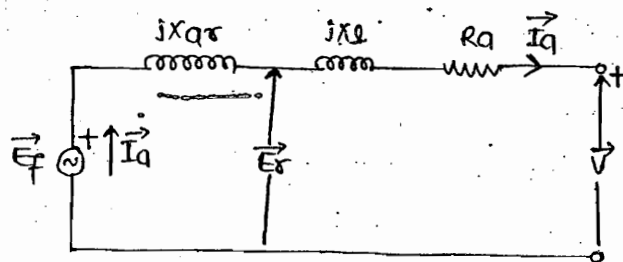
$$* \vec{E}_f = \vec{V} + \vec{I}_a R_a - \vec{E}_{al} - \vec{E}_{ar} \text{ ----- (i)}$$

$$= \vec{V} + \vec{I}_a R_a + j \vec{I}_a X_l + j \vec{I}_a X_{ar}$$

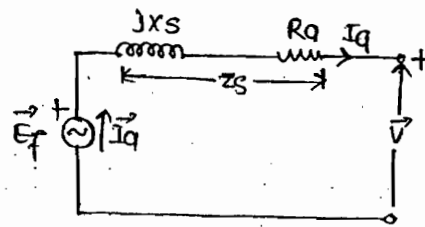
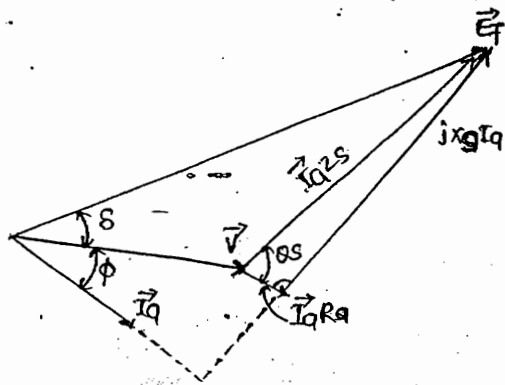
$$= \vec{V} + \vec{I}_a R_a + j \vec{I}_a (X_l + X_{ar})$$

$$= \vec{V} + \vec{I}_a R_a + j \vec{I}_a X_s \text{ ----- (ii)}$$

* equivalent ckt →



eqⁿ (i) ckt diagram



eqⁿ (ii) eq. ckt

Angle between F_f & ϕ_r = torque angle
 angle between $\vec{E}_f$ & $\vec{V}$ = power angle / Rotor angle.

air gap in 200MW
 syn. m/c is 70mm
 500MW - 130mm

Open circuit Test
 &
 Short circuit Test

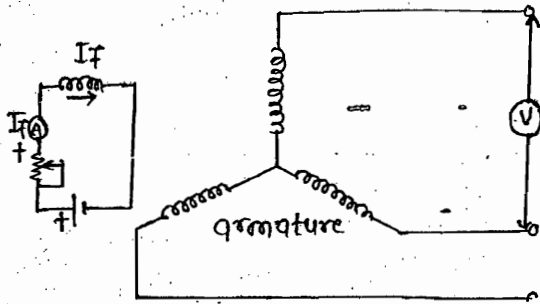


Fig-oc test

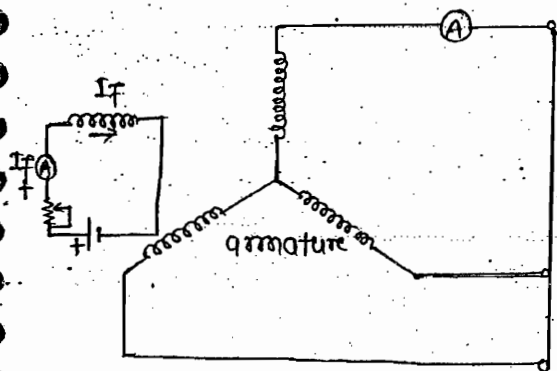
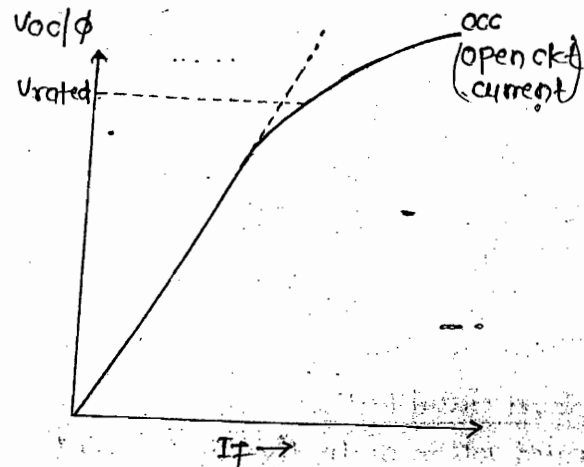
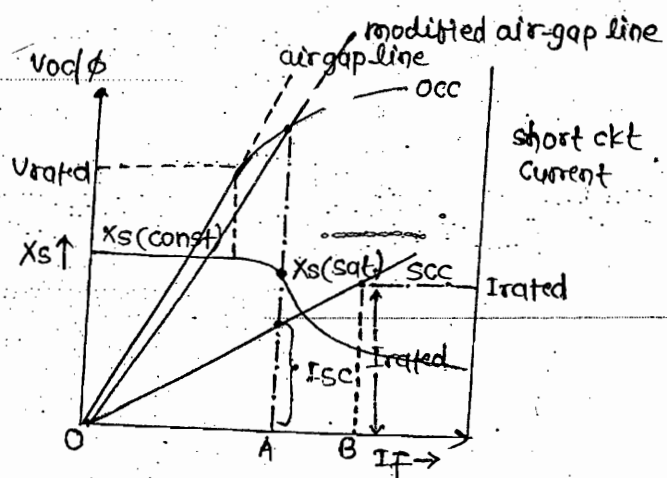


Fig-sc test



* allow 1.2-2 times of FL current.

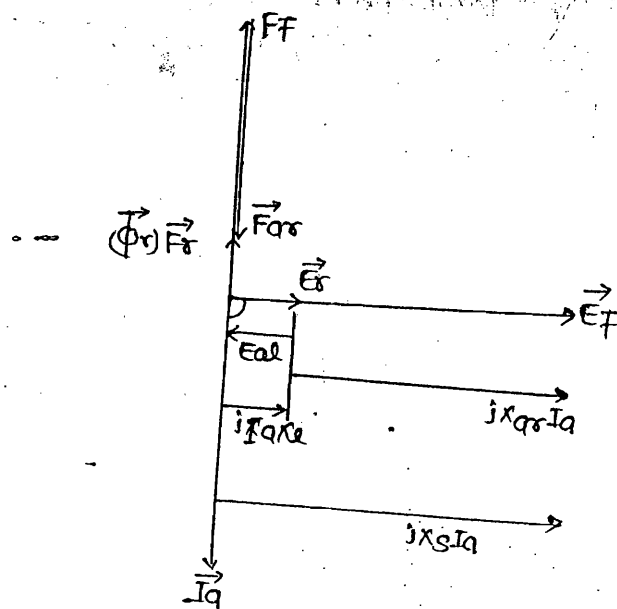
(X_s constant at 1st then variable)

* modified air gap line is the occ of an eq. unsaturated m/c that gives rated vol. at the same field current as the actual m/c.

* I_{sc} by definition is that value of sc arm. current which is obtained at a field current that gives rated vol. on oc.

$$X_s(sat) = \frac{V_{rated}}{I_{sc}}$$

$X_s(sat)$ = Saturated synchronous Reactance (or) Adjusted syn. reactance



* Short circuit Ratio is defined as the ratio of the field current req. to give rated vol. on ac to the field current req. to give rated arm. current on sc.

Short circuit Ratio, $SCR = \frac{OA}{OB}$

$$SCR = \frac{I_{sc}}{I_{rated}}$$

I_{rated} = Base value, I_{sc} = actual value

$$\frac{I_{sc}}{I_{rated}} = I_{pu} = SCR$$

$$= \frac{V_{rated}/X_s(sat)}{I_{rated}}$$

$$\therefore X_s(sat) = \frac{V_{rated}}{I_{sc}}$$

$$= \frac{V_{rated}/I_{rated}}{X_s(sat)}$$

$$= \frac{Z_{base}}{X_s(sat)}$$

$$SCR = \frac{1}{X_s(sat)/Z_{base}}$$

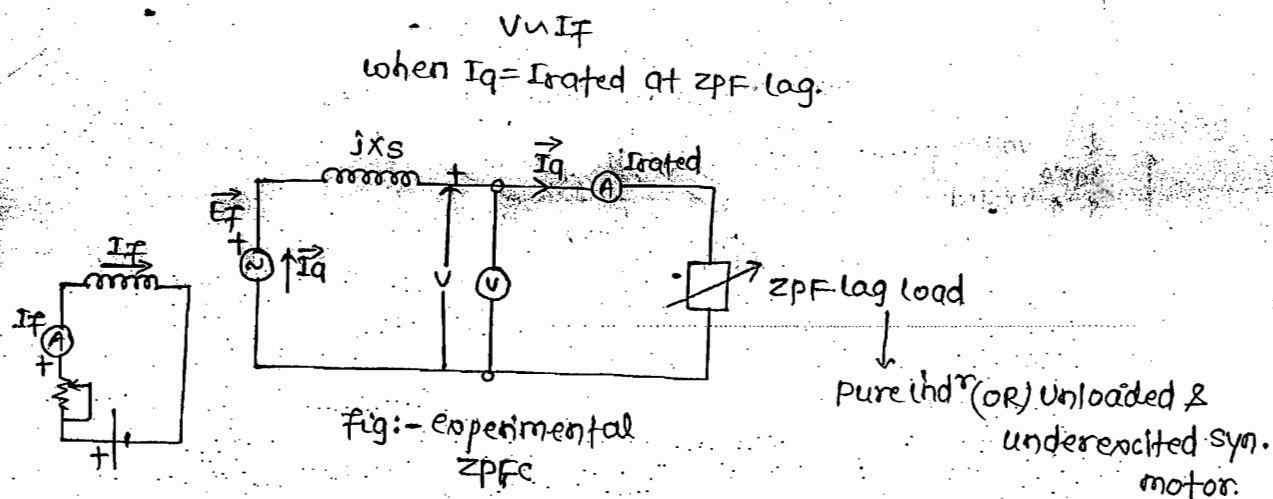
$$SCR = \frac{1}{X_s(sat)_{pu}}$$

(i) V_{t} will be also increases because high value of vol. drop.
 (ii) Stability decreases.

- * For turbo gen^r 0.58 but not less than 0.35 (By IEEE^{For} SCR)
- * Salient pole gen^r the client needs (or) ~~is~~ not less than 0.8.

Zero Power Factor ZPF c/s (OR) Potier c/s

* ZPF c/s, also called Potier c/s shows the variation of terminal vol. against field current when arm. current is maintained at its rated value & ZPF lag.



* Since the variation in field current (or) the variation in load affects the arm. current as well as the terminal vol., it is recommended that the field current be increased for terminal vol. & subsequently the load may be adjusted for rated arm. current.

I_f	V	I_a
3.2A	225V	200A
3.5A	246V	218A
3.5A	257V	200A

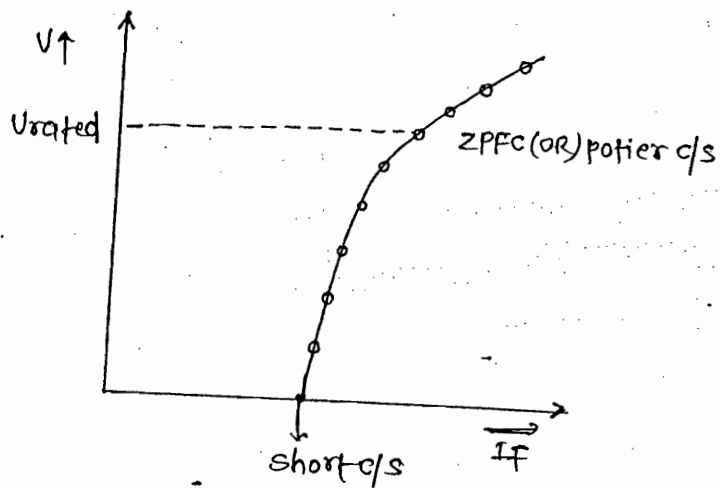
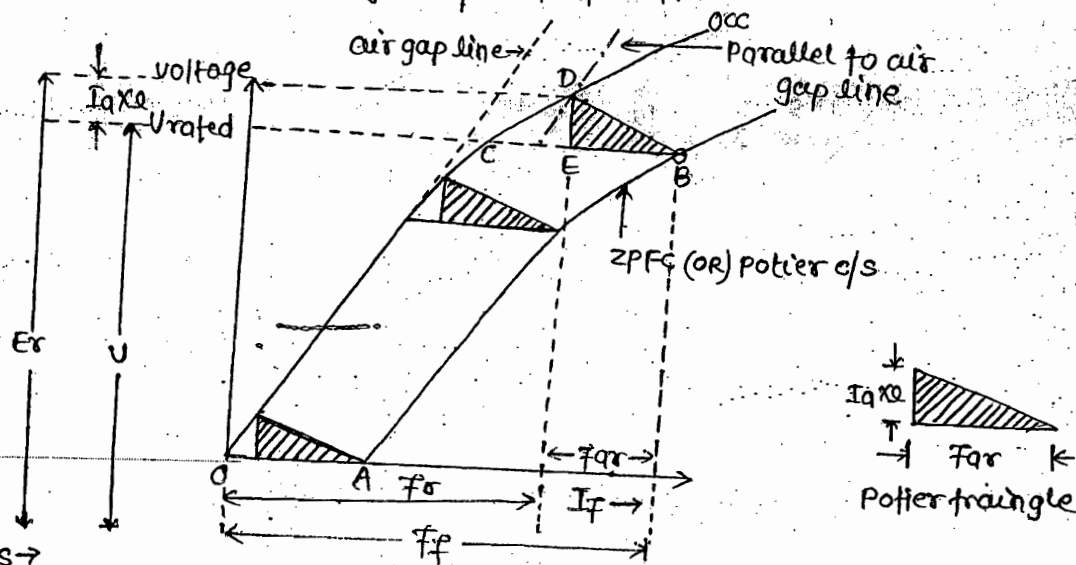


Fig:- Experimental ZPFC



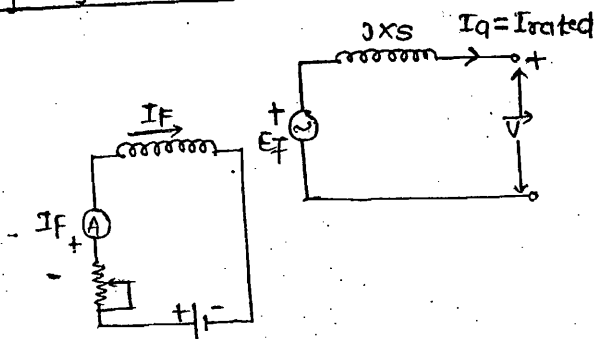
Steps →

- * Draw the OCC & locate the 2 experimental points A & B. point A represents the field current req. to give rated arm. current on sc. point B represents
- * point the field current req. to give rated terminal vol. when the arm. current equals rated value at zpf lag.
- * Take $BC = OA$ as shown in fig.
- * From point C, draw a line parallel to the air gap line intersecting the OCC at point D.
- * From point D, draw a perpendicular DE on BC. The ΔDEB is the potier Δ . In the potier Δ BE represents the effect of F_{ar} in the air gap. DE represents the arm. leakage reactance drop on FL & its called potier drop.

* move the potier Δ parallel to itself while always keeping its vertex D on the ooc. The locus of point B is the zpfc also called potier q/s.

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* Voltage Regulation $\rightarrow$



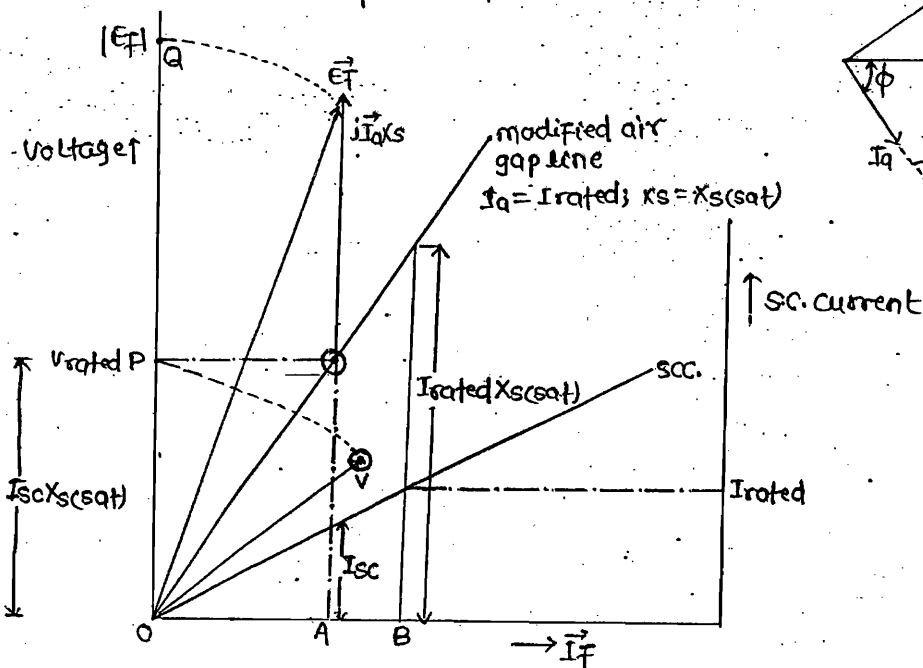
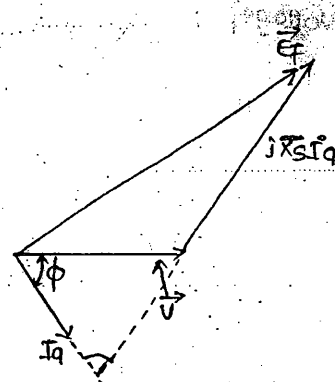
$$\text{Voltage Reg}^n = \frac{E_f - V}{V} \text{ PU}$$

where $V = V_{rated}$.

* UR^n of an alternator is defined as the rise in o/p terminal vol. expressed as a fraction of FL rated vol. when FL at a specified pf is thrown off keeping the excitation constant.

UR^n by EMF method $\rightarrow$

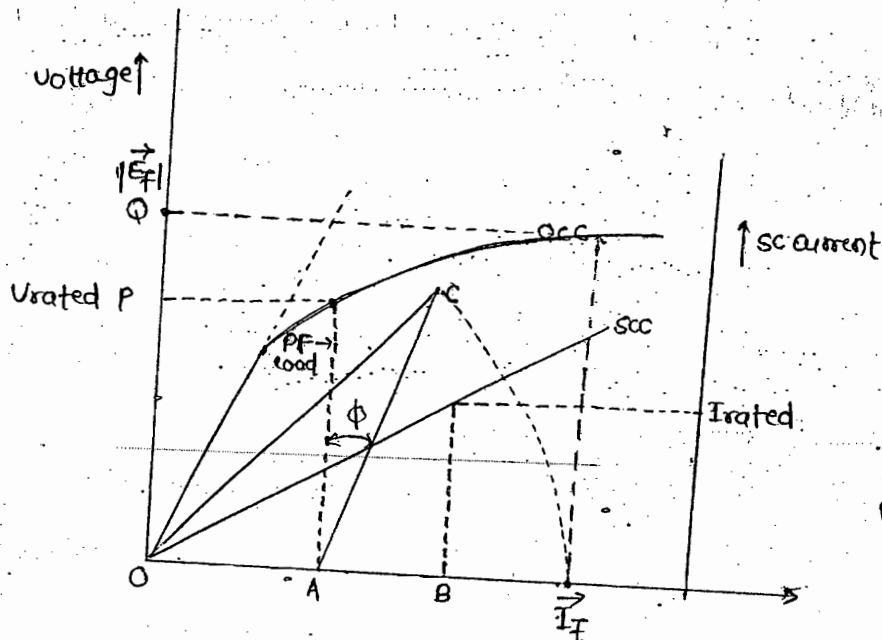
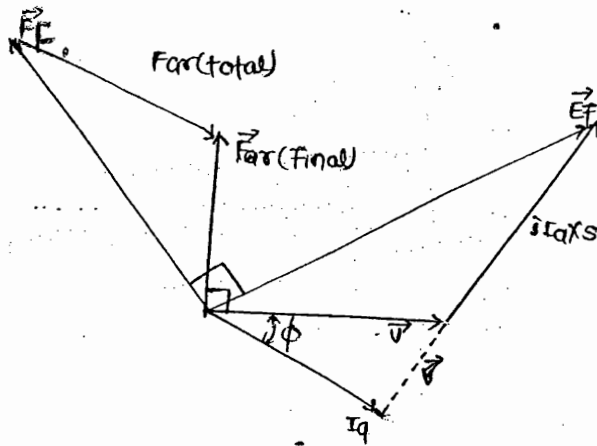
$$\vec{E}_f = \vec{V} + j\vec{I}_a X_s$$



$$UR^n = \frac{OQ - OP}{OP} \text{ PU}$$

* The value of UR^n obtained by EMF method is higher than actual & therefore this method is called pessimistic.

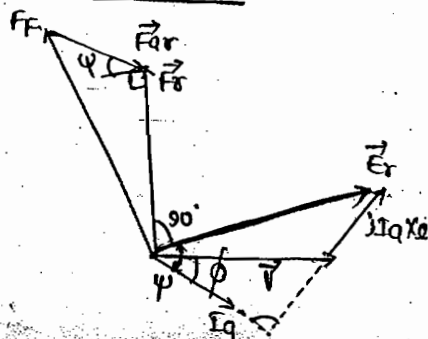
* V_R^n by mmf method $\rightarrow$ (OR) amp-turn method.

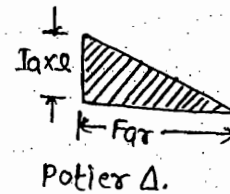
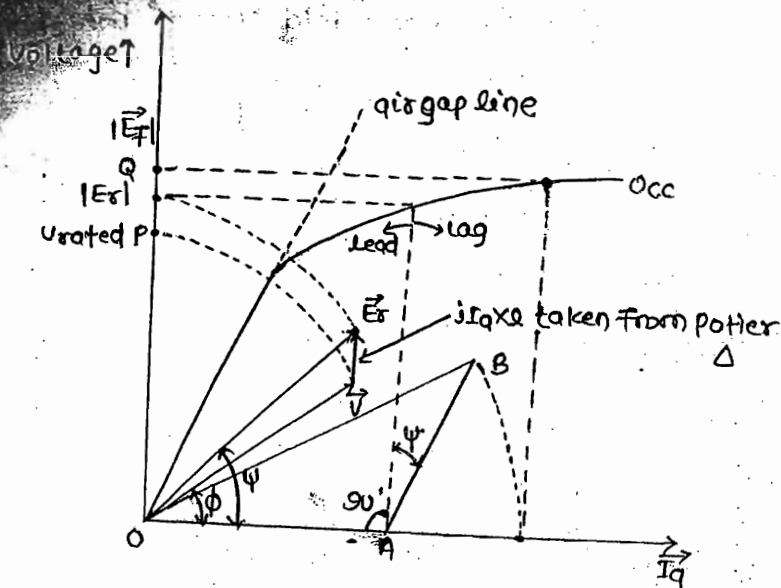


$$\begin{aligned} OA &\equiv F_r(\text{Final}) \\ OB &\equiv F_{ar}(\text{total}) \\ AC &= OB \\ \therefore OC &\equiv F_f \\ V_R^n &= \frac{OQ - OP}{OP} \end{aligned}$$

* The value of the V_R^n obtain by mmf method is less than actual & therefore this method is called optimistic method.

V_R^n by ZPF method (OR) potier method $\rightarrow$





* The V_R obtained by EMF method was found to be pessimistic while the mmf method produced the optimistic.

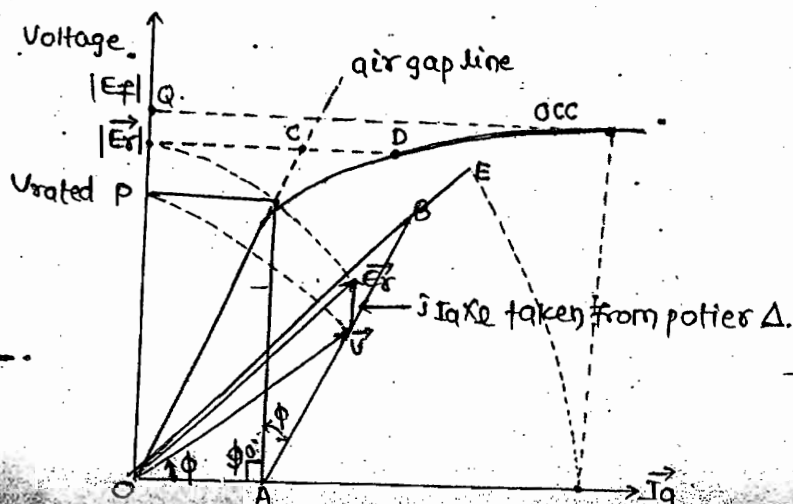
The Potier method also called ZPF method treats the quantity as they actually exist in a m/c on load.

The V_R obtained by ZPF method therefore more realistic than the earlier 2 methods.

Thus the potier method was long used for the determination of V_R of syn. gen^r until American std. association (ASA) proposed their modification of mmf method.

This method is now popularly known as ASA method.

* V_R by ASA method $\rightarrow$



* 1pu field current in a syn. m/c is that value of field current which gives rated voltage on the air gap line.

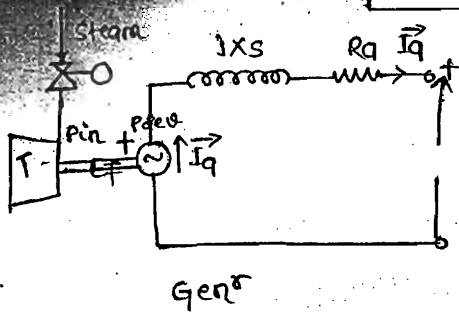
* The ASA modified the mmf method by determining F_r final from the air gap line instead of from the OCC. It was then combined with F_{ar} total taken from SC data to determine the F_f of unsaturated m/c.

* Since the degree of saturation depends on the air gap mmf, the additional field mmf req. to overcome saturation was determined as a horizontal intersect b/n the air gap line and OCC at E_f level.

* This additional field mmf was added to the F_f of unsaturated m/c to finally determine the F_f of actual m/c under actual operating condn.

* The ASA method has been found to give satisfactory method for cylindrical rotor & as well as the salient pole m/c & it is latest recommendation for determination of V_{rn} of syn. gen^r.

POWER ANGLE RELATIONS



$$\vec{E}_f = \vec{V} + \vec{I}_q Z_s$$

$$\vec{I}_q = \frac{\vec{E}_f - \vec{V}}{Z_s} = \frac{E_f \angle \delta - V \angle 0^\circ}{Z_s \angle \theta_s}$$

$$\vec{I}_q = \frac{E_f \angle \delta - \theta_s - \frac{V \angle -\theta_s}{Z_s}}$$

$$S_{out} = \vec{P}_{out} + j \vec{Q}_{out}$$

$$= V \vec{I}_q^*$$

$$= V \angle 0^\circ \left[\frac{E_f \angle \theta_s - \delta}{Z_s} - \frac{V \angle \theta_s}{Z_s} \right]$$

$$S_{out} = \frac{V E_f \angle \theta_s - \delta}{Z_s} - \frac{V^2 \angle \theta_s}{Z_s} \quad \text{--- (i)}$$

$$\therefore P_{out} = \frac{V E_f}{Z_s} \cos(\theta_s - \delta) - \frac{V^2}{Z_s} \cos \theta_s$$

P_{out} is max^m when $\delta = \theta_s$ (variable = constant)

$$\text{then } P_{out}(\max) = \frac{V E_f}{Z_s} - \frac{V^2}{Z_s} \cos \theta_s$$

$$\vec{S}_{dev} = P_{dev} + j Q_{dev} = \vec{E}_f \vec{I}_q^*$$

$$= E_f \angle \delta \left[\frac{E_f \angle \theta_s - \delta}{Z_s} - \frac{V \angle \theta_s}{Z_s} \right]$$

$$\vec{S}_{dev} = \frac{E_f^2}{Z_s} \angle \theta_s - \frac{V E_f}{Z_s} \angle \theta_s + \delta \quad \text{--- (ii)}$$

$$P_{dev} = \frac{E_f^2}{Z_s} \cos \theta_s - \frac{V E_f}{Z_s} \cos(\theta_s + \delta)$$

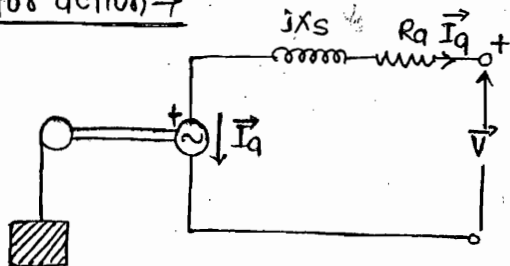
P_{dev} is max^m when -

$$\theta_s + \delta = 180^\circ$$

$$\delta = (180 - \theta_s)$$

And this decides steady state stability limit.

motor action →



$$E_f = V - \vec{I}_a Z_s$$

$$\vec{I}_a = \frac{V - E_f}{Z_s}$$

$$\vec{I}_a = \frac{V \angle 0^\circ - E_f \angle -\delta}{Z_s \angle \theta_s}$$

$$\vec{I}_a = \frac{V \angle -\theta_s - E_f \angle -(\theta_s + \delta)}{Z_s}$$

$$S_{dev} = P_{dev} + jQ_{dev}$$

$$= E_f \vec{I}_a^*$$

$$= E_f Z_s \left[\frac{V \angle \theta_s - E_f \angle (\theta_s + \delta)}{Z_s^2} \right]$$

$$S_{dev} = \frac{VE_f \angle \theta_s - E_f^2 \angle 0^\circ}{Z_s} \quad \text{--- (3)}$$

$$\text{Now; } P_{dev} = \frac{VE_f}{Z_s} \cos(\theta_s - \delta) - \frac{E_f^2}{Z_s} \cos \theta_s$$

P_{dev} is max^m when $\theta_s = \delta$ & this decides steady state stability limit of the motor.

$$\& \quad P_{dev}(\max) = \frac{VE_f}{Z_s} - \frac{E_f^2}{Z_s} \cos \theta_s$$

$$S_{in} = P_{in} + jQ_{in}$$

$$= E \cdot \vec{V} \cdot \vec{I}_a^*$$

$$= V \left[\frac{V \angle \theta_s - E_f \angle (\theta_s + \delta)}{Z_s} \right]$$

$$S_{in} = \frac{V^2 \angle \theta_s - VE_f \angle (\theta_s + \delta)}{Z_s} \quad \text{--- (4)}$$

$$P_{in} = \frac{V^2}{Z_s} \cos \theta_s - \frac{V \cdot E_f}{Z_s} \cos(\theta_s + \delta)$$

Neglecting $R_a \rightarrow$

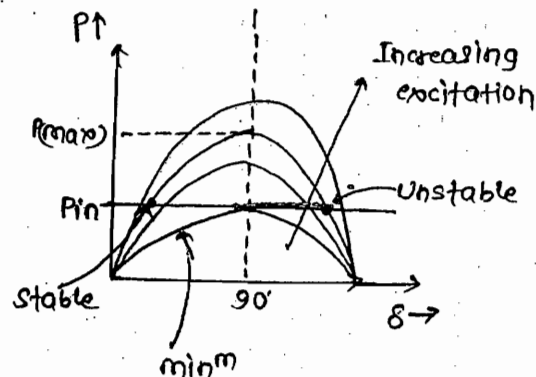
$R_a = 0$; then $Z_s = X_s, \theta_s = 90^\circ$

Gen^r $\rightarrow P_{out} = P_{dev} = P = \frac{V \cdot E_f \cdot \sin \delta}{X_s}$

motor $\rightarrow P_{dev} = P_{in} = P = \frac{V \cdot E_f \cdot \sin \delta}{X_s}$

$\& P_{max} = \frac{E_f V}{X_s}$ at $\delta = 90^\circ$

$P = P_{max} \sin \delta$



* Reactive Power $\rightarrow$

Gen^r $\rightarrow$

$$Q_{out} = \frac{V}{X_s} (E_f \cos \delta - V)$$

motor $\rightarrow$

$$Q_{in} = \frac{V}{X_s} (V - E_f \cos \delta)$$

Gen^r $\rightarrow$

Case (i) $\rightarrow$

when $E_f \cos \delta = V$ i.e. normally excited gen^r then $Q_{out} = 0$.

And therefore operating at UPF.

case (ii) $\rightarrow$

when $E_f \cos \delta$ is more than V i.e. over excited gen^r then $Q_{out} = +ve$

i.e. supplying lagging VARS. And therefore operating at lagging PF.

case (iii) $\rightarrow$

when $E_f \cos \delta$ is less than V i.e. under excited gen^r, then $Q_{out} = -ve$

i.e. supplying leading VARS & therefore operating at leading PF.

motor $\rightarrow$

case (i) $\rightarrow$

when $E_f \cos \delta = V$; i.e. normally excited gen^r then $Q_{out} = 0$

& therefore operating at UPF.

Case (ii) $\rightarrow$

when $E_f \cos \delta > V$ i.e. $Q_{out} = -ve$ i.e. taking leading VARS & $\therefore$ operating at leading VARS.

Case (3) →

When $E_f \cos \delta < V$; i.e. underexcited motor then $Q_{in} = +ve$ i.e. absorbing lagging VAR & ∴ operating at lagging PF.

* Effect of change in excitation at constant (kw) o/p →

$$P = VI_a \cos \phi$$

$$\propto I_a \cos \phi$$

constant

$$P = \frac{VE_f}{X_s} \sin \delta$$

$$= E_f \sin \delta$$

(constant)

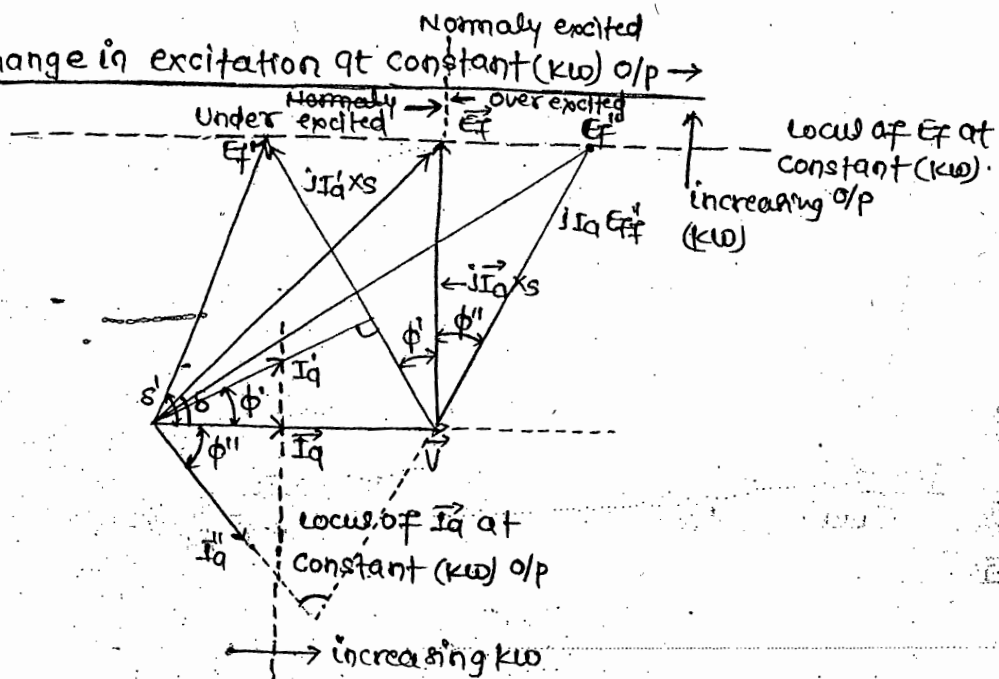


Fig → Gen^r

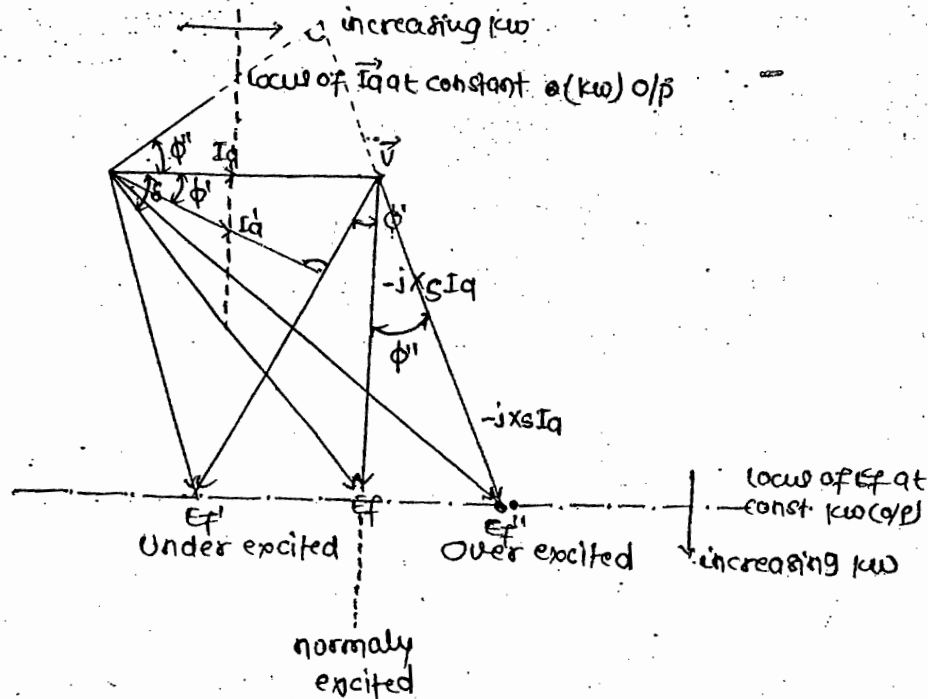


Fig → motor

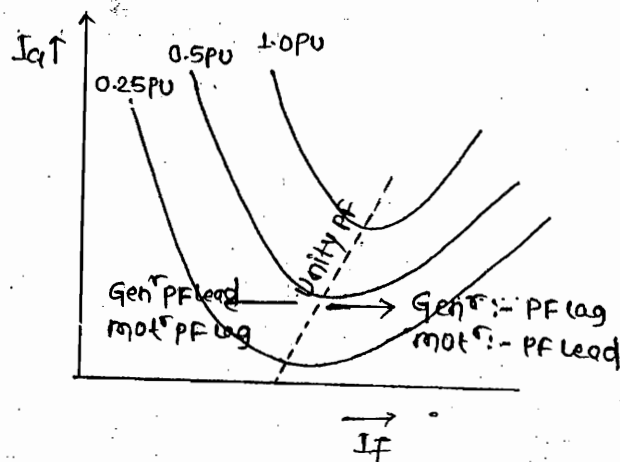


Fig:- Syn. m/c 'v' curves.

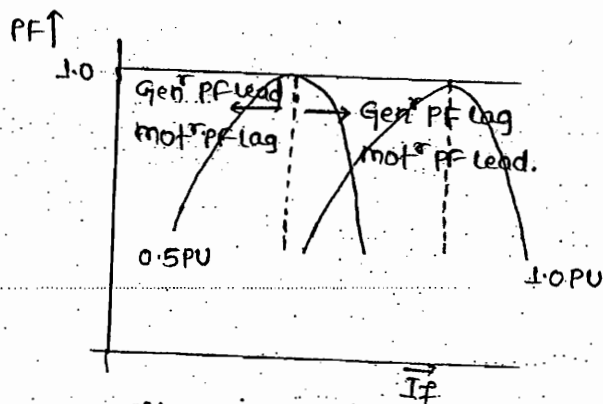
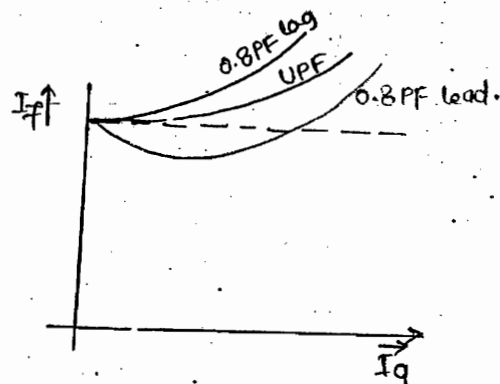


Fig:- syn. m/c inverted 'v' curves.

* Compounding curve of a gen^r shows the variation of the field current req. to maintain constant terminal vol. when a const. PF load is varied.



Gen^r compounding curves.